INCH-POUND
MIL-D-82586B(OS)
31 May 1990
SUPERSEDING
MIL-D-82586A(OS)
1 February 1974

MILITARY SPECIFICATION

DIRECTOR, GUN AND GUIDED MISSILE, MK 73 MOD 2 AND ASSOCIATED EQUIPMENT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* **1.1 Scope.** This specification covers the performance and construction requirements for Director, Gun and Guided Missile, Mk 73 Mod 2 and associated equipment, which together comprise one of the main elements of Guided Missile Fire Control System Mk 74 Mod 14.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

* MMM-A-1617	Adhesive, Rubber Base, General Purpose

MILITARY

MIL-P-116	Preservation, methods of
MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1210

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	MIL-G-12803	Gasket Material, Non-metallic
*	MIL-E-17555	Electronic And Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of
	MIL-F-18870	Fire Control Equipment, Naval Shipboard, General Specification For
	MIL-I-26860	Indicator, Humidity, Plug, Color Change

STANDARDS

FEDERAL

FED-STD-151	Metals: Test Methods

MILITARY

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the control of Electromagnetic Interference
MIL-STD-471	Maintainability Verfication/Demonstration/Evaluation
MIL-STD-740	Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-781	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-1399-300	Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating Current (METRIC)
* MIL-STD-2073-1	DOD Material Procedures for Development and Application of Packaging Requirements

HANDBOOKS

MIL-HDBK-217	Reliability Prediction of Electronic Equipment
MIL-HDBK-472	Maintainability Prediction

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL SEA SYSTEMS COMMAND

2805657G1	Director, Gun and Guided Missile, Mk 73 Mod 2
2805684G1	Ring Electrical Contact Mk 4 Mod 1
2805685G1	Controller, Director, Mk 26 Mod 0
2805687G1	Relay Panel Mk 321 Mod 0

(Application for copies of NAVSEA drawings should be addressed to: Commanding Officer, Naval Ordnance Station, Attn: Code 802, Louisville, KY 40214.)

* **2.2 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Equipment definition. Director, Gun and Guided Missile, Mk 73 Mod 2, shall obtain the performance specified herein when operated with associated equipments as follows:

Equipment	Drawing
a. Director, Gun and Guided Missile, Mk 73 Mod 2	2805657G1
b. Controller, Director Mk 26 Mod 0	2805685G1
c. Ring Electrical Contact Mk 4 Mod 1	2805684G1
d. Relay Panel Mk 321 Mod 0	2805687G1

Director, Gun and Guided Missile, Mk 73 Mod 2 together with the above listed equipments, for the purposes of this specification may be hereinafter referred to as simply "the director."

3.2 Performance. The director shall mount the antenna assemblies for Radar Set AN/SPG-51D. Rate control of the director shall be in response to rate orders provided to the director. The director shall be capable of motion in two axes, train and elevation (parallel to and normal to base plane of the director) and shall be unmanned with start, stop, and reset controls remotely located. The director shall not be limited in train and traverse positioning. Positioning at deck elevation angles from +83 degrees to −30 degrees is required. The director shall supply director train and elevation position data for use by Guided Missile Fire Control System Mk 74.

3.2.1 Modes of operation. The director shall operate as a subsystem in the Mk 74 Fire Control System. The director shall release its brakes and shall be positioned in response to ordered velocity signals when the Director Assigned signal is present. The removal of the Director Assigned signal shall set the train and elevation parking brakes. Thus the basic fire control system modes of Air Ready, Designate, and Track shall be capable of being accommodated by the director. Specific Director modes are described below.

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3.2.1.1 Start Up Complete. The director shall operate in the Start Up Complete mode within 15 seconds of the receipt of a 50 volts direct current (Vdc) signal if all required power is available and the interlocks are closed. During this mode, the director shall be completely energized except no excitation shall be applied to the train and elevation motor armatures. The director shall be restrained from motion in both axes by means of brakes. Director start up must be complete before the director can be switched into the other modes.

*

3.2.1.2 Designation. In response to fire control system command inputs, the director shall slew simultaneously in traverse and elevation at rates up to and including 1 rad/s within the limits set forth herein. Within the limits of the power drive servo requirements, the director shall also respond to designation search pattern commands, as defined by tables I and II and figures 1 and 2.

3.2.1.3 Director Assigned. In the Director Assigned mode, the director shall be positioned in response to ordered rate signals, designated bearings (DB's) and designated elevations (DE's).

3.2.1.3.1 Track. With maximum normal deck motion the director shall follow angular space rate orders up to and including 12 degrees per second up to a deck elevation of 70 degrees. With no ships motion the director shall follow angular rate orders up to and including 12 degrees per second relative to the director up to a deck elevation of 83 degrees.

3.2.1.3.2 Horizon search. The director shall follow a 35 degree/s traverse rate order at a true elevation of 11 milliradians (mrad) with a duty as specified herein.

3.2.1.4 Director Test. The director shall operate in the Director Test mode upon receipt of a 50 Vdc Director Test signal. In this mode the director shall not respond to Director Assigned or Battle Short signals or to rate orders from the fire control system and shall only respond to signals from the test panel of the Controller. The director train and elevation power drives may be tested separately or simultaneously.

3.2.1.5 Battle Short. The director shall operate in th Battle Short mode upon receipt of a 50 Vdc Battle Short signal. In the Battle Short mode, director shutdown due to motor overloads, drive overspeed, electronics overtemperature, hydraulic power supply overload, or loss of motor cooling air is prohibited.

3.2.2 Train and elevation power drive servo requirements. The director shall follow space angular rates, deck angular rates, and angular accelerations up to and including the following:

*

Axis	Deck Angular Rate	Space Angular Rate	Angular Acceleration
Traverse	2.2 rad/s	1 rad/s until train deck rate is exceeded[1]	2 rad/s^2 at 0° Ed' 1.4 rad/s^2 at 65° Ed'
Train			2 rad/s^{2}at 0° Ed' 3.3 rad/s^2 at 65° Ed'
Elevation		1 rad/s	2.25 rad/s^2

[1]Train rotation = Traverse Rotation × seconds Ed' with no ship's motion.

The elevation angular rate requirement shall be met for deck evaluation angles from +73 degrees to −20 degrees. The elevation angular rate shall decrease linearly from 1 rad/s for deck elevation angles from +73 degrees to +83 degrees and −20 degrees to −30 degrees.

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TABLE I. *Search coverage in designate modes.*

Mode	Required angle	Search program	Angle search pattern	Time per scan (s)	Time from director 1 Synch cumulative (s)
	Note 1		Note 2		
Air Target - NTDS Normal	A_2	a) 1 scan	E-rel	2	3
		b) 3 scans	E-rel	2	9
Air Target - NTDS Elevation unreliable	A_3	a) 1 scan	E-rel	2	3
		b) 3 scans	E-rel	2	9
		c) 3 scans	E-unrel	11	43 max
Air Target - TDT	A_1	a) 2 "looks"			1.5
		b) 3 "looks"			2.25
Air Target - IDD	A_1	a) 1 "look"			1.25
		b) 3 "looks"			2.0
Air Target - NTDS Passive Acquisition	A_2				
Surface Target Any Source	A_4	+5° Scan in By	Surface	1.6	2.1
Local Designate	Same As comparable NTDS designate				

NOTES:

1) Required Angle Coverage A is:

A_1 within 0.8° of designated point (no search)
A_2 within 2.0° of designated point
A_3 +2° in bearing; Elevation reliable from 0° E to 45° E or 100k ft; elevation unreliable, see figure 2
A_4 +5° in bearing with 1.6 s period; elevation fixed at 11mr above the surface

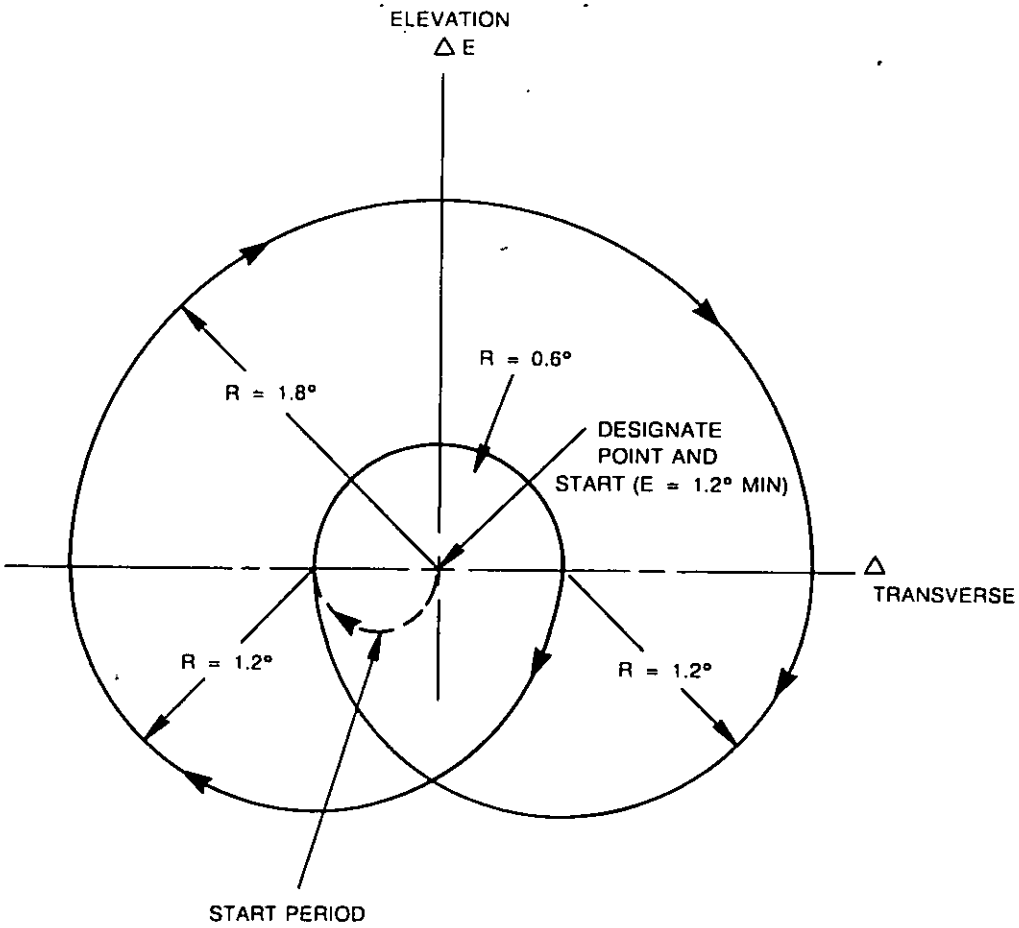
2) Angle Search Pattern: E-rel, see figure 1; E-unrel, see figure 2.

TABLE II. *Director motion required for search coverage.*

Required coverage	Path	Space Rad. (deg)	Period (s)	Max space rate (deg/s)	Max space accel (deg/s^2) traverse & elevation	Ed' max. (deg)	Max train vel (deg/s)	Max train accel (deg/s^2)	Max elev vel (deg/s)	Max elev accel (deg/s^2)
A_2	Inner	0.6	1	3.8	24	65	9	57	3.8	24
	Transfer	1.2	1	7.6	48	65	18	115	7.6	48
	Outer	1.8	1	11.4	71	65	27	167	11.4	71
A_3	See figure 2									
A_4	El	±0	1.6							
	Trav	±5		20	77	20	21	82		

NOTE: Angle coverage A_2 and A_3 as shown in figures 1 and 2.

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PERIOD:	2 ± 0.2 s
MAX TRANSVERSE ACCELERATION:	≤72°/ s^2
MAX ELEVATION ACCELERATION:	≤100°/ s^2
COVERAGE SHOWN FOR d(Ed') ≤65°.	REDUCED COVERAGE FOR d(Ed') >65°.

FIGURE 1. *Antenna search pattern in plane normal to the line of sight elevation reliable designations.*

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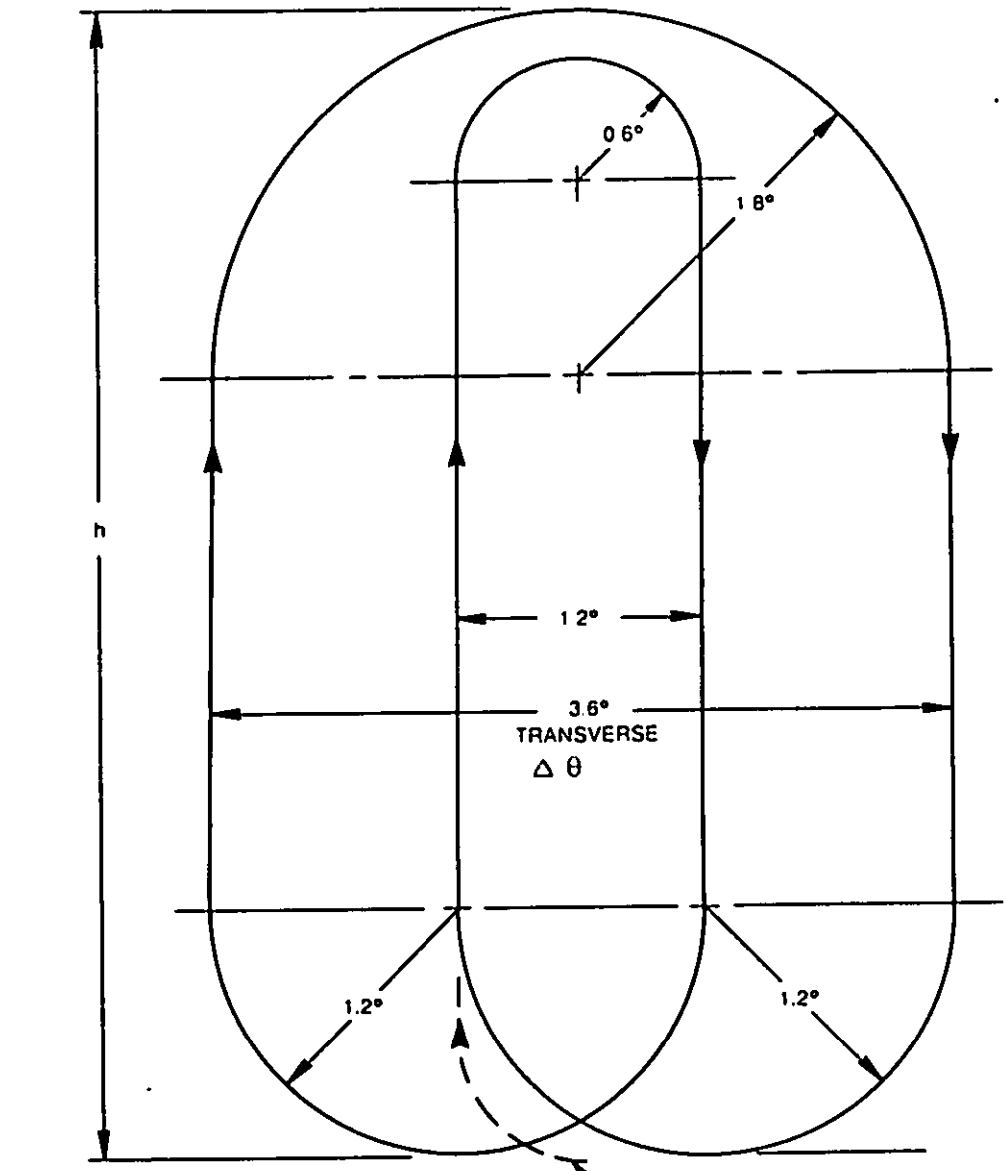


FIGURE 2. *Antenna search pattern in plane normal to LOS Elevation Unreliable designations.*

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* **3.2.2.1 Frequency response.** The closed loop frequency response of the director traverse and elevation servos from space rate input to space rate output shall be consistent with the specified requirements herein. In addition, the amplitude ratio of the output to input over the frequency range from 14 to 180 Hz shall not exceed the following limits:

Frequency Range	*Output/Input (Magnitude)*
14 Hz to 28 Hz	$(14/f)^2$ (O dB to −12 dB)
28 Hz to 180 Hz	Equal to or less than −12 dB

Any peaking of the amplitude ratio below 14 Hz shall not exceed 4.45 decibels (dB). 0 dB shall be referenced to an amplitude ratio of 0.1 Hz.

All frequency response requirements shall apply for space rate commands which lie in the range of 5 to 70 percent of the full scale input valve, produce outputs which lie within the position, rate, and acceleration capability of the director, and have an Ed' range of −20 to +73 degrees.

3.2.2.2 Response to step input space rates. The director shall respond to steps in ordered space rates from −70 to +70 to −70 percent of the space rate requirements specified herein without sustained oscillation. The traverse requirements shall apply for deck elevation angles from −20 degrees to +60 degrees.

3.2.2.3 Base motion isolation in absence of wind. The director shall isolate the LOS from the influence of ship's motion such that the filtered velocity error due to ship's motion (exclusive of the effects of relative wind) shall not exceed 0.17 mrad/s in elevation. With Ed' of 60 degrees or less, filtered velocity error due to ship's motion shall not exceed 0.17 mrad/s in traverse. The above filtered velocity errors shall be measured at the output of a filter whose transfer function is:

$$H(S) = \frac{1}{4S^2 + 4S + 1}$$

3.2.2.4 Response to wind torques. The director shall isolate the LOS from the influence of wind (a maximum of 75 knot relative wind) such that the static measured velocity error due to static wind shall not exceed 0.2 mrad/s (with the antenna in a tracking position to cause maximum torque). When measured at the output of the filter specified above, the varying velocity error due to ship's motion effect on the relative wind shall not exceed 0.2 mrad/s and the varying velocity error due to wind gustiness shall not exceed 0.4 mrad/s.

3.2.2.5 Servo loop summing accuracy. The steady state response signal from the gyro shall be equal to the static rate command within 0.25 mrad/s plus 0.1 percent of the rate command from zero to full scale rate command. This deviation does not include gyro subsystem errors, errors due to base motion, or errors due to wind torque.

3.2.2.6 Rate measuring accuracy. Gyro subsystem rate measuring accuracy of the traverse, elevation, and cross traverse rates of the director shall be within 1 mrad/s plus 1 percent of the director space rate from zero to 1 rad/s. Cross traverse rate measuring accuracy from 1 rad/s to 1.5 rad/s shall be within 2 percent of the director
* space rate, and from 1.5 rad/s to 2.0 rad/s, shall be within 5 percent of the director space rate.

3.2.3 Overspeed limits. Overspeed sensors shall be provided to sense a director train rate of 156 degrees per second ±10 percent, and elevation rate of 90 degrees per second ±10 percent. When overspeed is sensed the related power drive shall be interrupted (power removed and brakes set).

3.2.4 Position limits. The director shall not be limited in train or traverse positioning. Mechanical limits shall be provided to restrict director elevation motion to angles between 88 degrees and −35 degrees inclusive.

3.2.5 Servo system duty.

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3.2.5.1 Train servo duty. The train power drive servo system duty shall be limited as indicated below:

Train Servo	Motor speed	Motor torque	Limited By
1 hr Duty	3.4 r/min	1500 ft lb	Thermal Overload
5 s	15 r/min	3300 ft lb	Thermal Overload
5 s	170 r/min	2000 ft lb	Motor Design
10 s	220 r/min	1000 ft lb	Motor Design
5 min	80 r/min	2000 ft lb	Motor Design

3.2.5.2 Elevation servo duty. The elevation power drive servo system duty shall be limited as indicated below:

Elevation servo	Motor speed	Motor torque	Limited By
1 hr Duty	3.4 r/min	750 ft lb	Thermal Overload
5 s	50 r/min	2800 ft lb	Thermal Overload
5 min	95 r/min	1000 ft lb	Motor Design

3.2.6 Data transmission.

3.2.6.1 Train. Instruments shall be provided in the director to be driven from the train axis as follows:

One Synchro Control Transmitter at one speed, 23CX4a, or equal.
One Synchro Control Transmitter at 36 speed, 23CX4a, or equal.
One Synchro Torque Transmitter at one speed, 15TX4a, or equal.
One adjustable Switch for use as Radiation Hazard Limit.

3.2.6.2 Train data accuracy requirements. The mechanical rotation of the one speed synchros shall be equal to director rotation about the train axis with maximum error of 15 minutes of arc exclusive of synchro error. The mechanical rotation of the 36 speed synchro shall be 36 times the director rotation about the train axis with a maximum error of 2 minutes of arc peak to peak, measured at one speed, including synchro error.

3.2.6.3 Elevation. Instruments shall be provided in the director to be driven from the elevation axis as follows:

One synchro control transmitter at one speed, 23CX4a, or equal.
One synchro control transmitter at 36 speed, 23CX4a, or equal.
One synchro torque transmitter at one speed, 15TX4a, or equal.
One Adjustable Switch for use as Radiation Hazard Limit.

3.2.6.4 Elevation data accuracy. The mechanical rotation of the one speed synchros shall be equal to director rotation about the elevation axis with maximum error of 15 minutes of arc exclusive of synchro error. The mechanical rotation of 36 speed synchro shall be 36 times the rotation of the director about the elevation axis with a maximum error of 2 minutes of arc peak to peak, measured at one speed, including synchro error.

3.2.7 Interface. The director shall interface with the fire control system as indicated in table III. Detailed interface and interconnection summary listing for Fire Control System will be furnished as specified in the contract or order (see 6.2).

*

TABLE III. *Director - Fire control system interface.*

Functional signal	Polarity	Scale	Voltage max.	Supplied by	Supplied to
Rate order Traverse	CW Positive	80 V/rad/s	80 Vdc	D/A Converter	Director
Rate order Elevation	UP Positive	80 V/rad/s	80 Vdc	D/A Converter	Director

Note: Train is clockwise (CW) when viewed from above the director.

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* **3.2.8 Director end usage.**

3.2.8.1 Tracking. The director shall be capable of operating as part of a position feedback loop as shown in the block diagram of figure 3.

3.2.8.1.1 Transient performance. The position loop shall have the following transient performance for a step input between 2 and 16 mrad with no wind or base motion disturbances, no initial input velocity or acceleration, and over an Ed' range of −20 to +73 degrees.

Position error settling time:

(a) $3/\sqrt{K_a} + t_1$ seconds, maximum, to settle within 15 percent of the input step value or ±0.75 mrad, whichever is larger.
t_1 = 0.5 second or $0.3/\sqrt{K_a}$, whichever is larger.

(b) $5/\sqrt{K_a} + t_2$ seconds, maximum, to settle within ±0.75 mrad.
t_2 = 0.5 second or $0.5/\sqrt{K_a}$, whichever is larger.

Peak position overshoot: 45 percent of input plus 0.78 mrad, maximum.

3.2.8.1.2 Steady-state performance. The position loop shall have the following steady-state performance $10/\sqrt{K_a}$ seconds after application of a step input between 2 and 16 mrad with no wind or base disturbance, no initial input velocity or acceleration, and over an Ed' range of −20 to +73 degrees.

Peak position error: Not exceeding ±0.75 mrad.

Peak velocity data error: Not exceeding ±0.75 mrad/s, exclusive of director servo and gyro offset errors.

3.2.8.2 Search, designation, slave. The director shall be capable of operation as part of a position feedback loop as shown in the block diagram of figure 4.

3.2.8.2.1 Transient performance. The position loop shall have the following transient performance for a step input between 0.5 and 5.0 degrees with no wind or base motion disturbances, no initial velocity or acceleration, and over an Ed' range of −20 to +73 degrees.

Position error settling time: 1.0 second, maximum, to settle within 3 percent of the final value or ±0.75 mrad, whichever is larger.

Peak position overshoot: 35 percent, maximum.

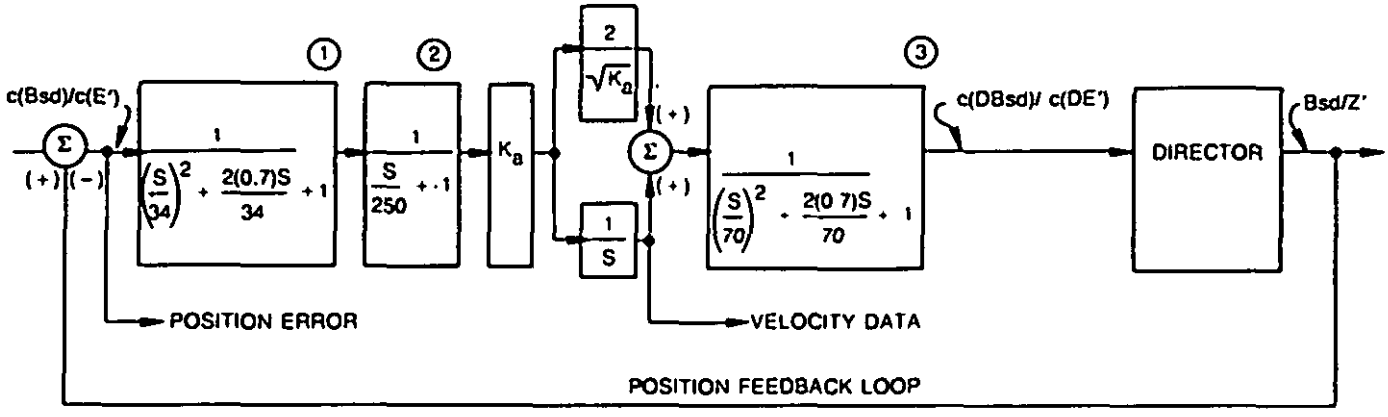
3.2.8.2.2 Steady-state performance. The position loop shall have a peak error not in excess of 0.75 mrad 3 seconds after application of a step input between 0.5 and 5.0 degrees with no wind or base motion disturbances, no input velocity or acceleration, and over an Ed' range of −20 to +73 degrees.

* **3.3 Physical Characteristics.**

3.3.1 Antenna group inertia. Performance shall be as specified herein when the inertia of the antenna group is as follows:

JE	285	±10 percent slug ft^2
JT @ −23 degree Ed'	301	±10 percent slug ft^2
JT @ 67 degree Ed'	217	±10 percent slug ft^2

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K_a = 1/64, 1/4, or 4/sec^2

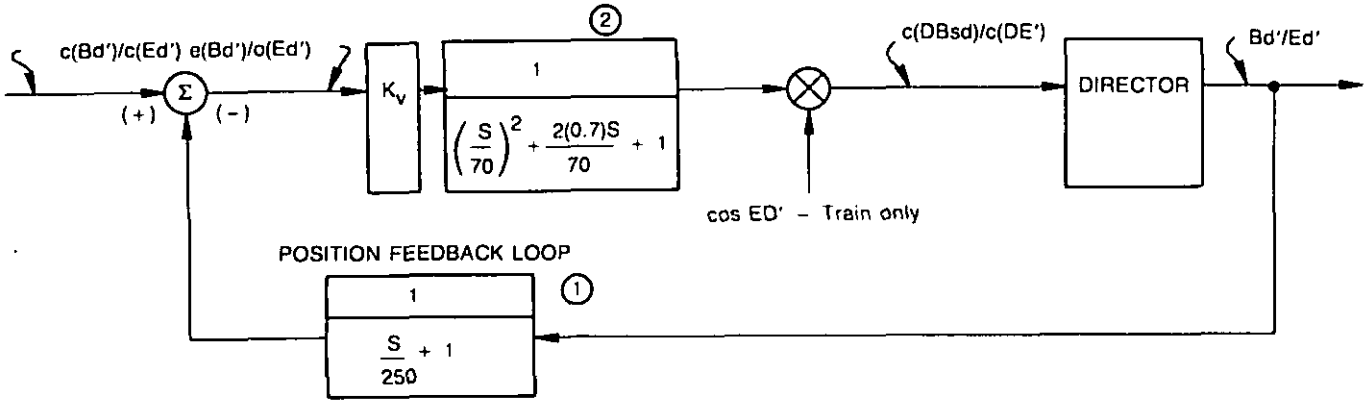
① - ANGLE ERROR FILTER

② - A/D CONVERTER FILTER

② - ANALOG REPRESENTATION OF DIGITAL TRANSPORT DELAY, UPDATE RATE, AND D/A CONVERTER

FIGURE 3. *Director track loop.*

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K_v = 7.0/sec

① – A/D CONVERTER FILTER

② – ANALOG REPRESENTATION OF DIGITAL TRANSPORT DELAY, UPDATE RATE, AND D/A CONVERTER

FIGURE 4. *Director designate, search and slave loop.*

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3.3.2 Antenna group center of gravity. For purposes of this specification, the center of gravity of that portion of the antenna group that is supported on the elevation axis shall be assumed to be at:

X = 27.3 inches ±5 percent
Y = 10.1 inches ±5 percent
Z = 5.85 inches ±5 percent

where the X-axis is defined as passing through the elevation axis at an angle normal to the antenna mounting surface.

3.3.3 Director moment of inertia. The nominal moment of inertia of the director, including counterweights for the antenna group, shall be:

JE	175	slug ft^2
JT @ −23 degree Ed'	670	slug ft^2
JT @ 67 degree Ed'	530	slug ft^2

3.3.4 Frictional loads. The frictional load of the director and slip ring assembly shall be as specified below. The director performance shall be as specified herein with the total frictional load as listed below.

a. Train axis	Director	80 ft lb, max
	Slip ring	50 ft lb, max
	Rotary joint	10 ft lb, max
	Total	140 ft lb, max
b. Elevation axis	Director	30 ft lb, max
	Rotary joint	10 ft lb, max
	Total	40 ft lb, max

3.3.5 Weights.

3.3.5.1 Train. Weight supported on the director train axis shall be as follows:

	Total	*Non-rotating*	*Rotating*
Slip ring ass'y	660 lbs	360 lbs	300 lbs
Director ass'y	7290 lbs	1900 lbs	5390 lbs
Antenna Group	1100 lbs	—	1100 lbs
Total:	9050 lbs	2260 lbs	6790 lbs

3.3.5.2 Elevation. Weight supported on the director elevation axis shall be as follows:

Antenna Group	778 lbs
Conterweights and arms	1160 lbs
Motor rotor	625 lbs
Other Director equipment	190 lbs
Total:	2753 lbs

3.3.5.3 Equipment. Weight of the controller, Director Mk 26 Mod 0 shall not exceed 1600 pounds. Weight of the Relay Panel Mk 321 Mod 0 shall not exceed 100 pounds.

3.3.5.4 Variations. Performance shall be as specified herein when the antenna group weight varies 10 percent. It shall be possible to install a TV camera or other equipment weighing up to 170 pounds and maintain balance by removing an equivalent amount of counterweight.

3.4 Construction requirements. The director shall be constructed in accordance with the drawings and specifications listed in Drawings 2805657G1, 2805684G1, 2805685G1, and 2805687G1, and as otherwise specified herein.

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3.4.1 Alignment.

3.4.1.1 Train bearing plane. The plane of the train bearing of the director shall be parallel to the train base of the director within ±1/2 mrad.

3.4.1.2 Elevation axis. The elevation axis of the director shall be parallel to the train base of the director within ±1/2 mrad.

3.4.2 Special mounting provisions. Mounting surfaces shall be provided for a boresight telescope and boresight television camera.

3.4.3 Stowing devices. Positive mechanical stowing shall be provided to prevent rotation of the director about the train and elevation axis under extreme service conditions and when director is off. These manually operated stow devices shall stow the director at a relative train angle of 180 degrees and the elevation axis at a relative elevation angle of 0 degrees with reference to the mechanical axis of the antenna reflector mounting surface. Change in the relative train angle of the reflector mounting surface in the stowed position shall be accomplished by rotating the entire director on its mounting surface and rezeroing the synchros. Interlocks shall be provided to prevent the application of power to the director drive motors when the director is stowed. No provision shall be made to by-pass these interlocks.

3.4.4 Brakes. Electrically controlled, hydraulically releasable brakes shall be provided to prevent rotation about the train and elevation axes under normal service conditions. The brakes shall automatically release upon the application of power to the director drive motors and be applied when drive power is removed. The brakes shall be manually releasable so that the director may be manually trained or elevated.

3.4.5 Manual drives. Manual drives shall be provided to position the director about its train and elevation axes. Manual hand wheels shall be positioned so that manual drive may be used in conjunction with manual brake release devices and train and elevation position indicators. Interlocks shall be provided to prevent the application of power to the director drive motors when the manual drives are engaged. No provision shall be made to by-pass these interlocks.

3.4.6 Cooling. The director and controller shall have a forced air prime mover internally located to provide cooling. For the purposes of this specification external ambient air drawn in at the base of the director for cooling shall not exceed 25 degrees Celsius (°C) and shall be filtered. The director will exhaust air at the base into the existing environment. The controller shall also exhaust into the existing environment.

3.4.7 Mounting. All equipment except the relay panel shall be base mounted. Equipment shall be designed such that it may be installed not less than 14 inches from above deck exterior walls except at vertical frame locations where the minimum mounting distance shall be 18 inches. For above deck interior compartment walls, the minimum mounting distance shall be 6 inches. The space behind equipments shall not be construed as access space.

3.4.8 Maximum dimensions. Maximum dimensions of equipment, assemblies or units intended for installation in above deck compartments shall be such as to permit passage through an opening 30 inches by 66 inches with a sill of 9 inches and 8 inch corner radii. For below deck installation the maximum dimensions shall be such as to permit passage through an opening 30 inches by 48 inches with 8 inch corner radii.

3.4.9 Degree of enclosure. Equipments installed in areas exposed to the elements shall be spraytight as specified in MIL-STD-108. Equipments located in enclosed areas shall be drip-proof (at a 15 degree angle) as specified in MIL-STD-108.

3.4.10 Accessibility. Access to all subassemblies and parts shall be from the front. Meters, dials, controls, or other shock sensitive equipment shall not be located on covers that must be removed during normal servicing.

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3.4.11 Pockets, wells, traps. Pockets, wells, traps, and the like in which water or condensed moisture can collect when the equipment is in operating or stowed position, shall be avoided or properly drained.

3.4.12 Cabling. Wiring or cabling furnished with the director and installed externally to the director as part of the ships wiring or cabling shall be in accordance with the applicable ship specifications.

3.4.13 Monitored parameters. The following parameters shall be monitored and displayed as indicated:

Function	Condition	Display Location
Ship's Power		
440 V, 60 Hz, 30	Voltage present	Director Controller
440 V, 400 Hz, 30	Voltage present	Director Controller
115 V, 60 Hz, 10	Voltage present	Director Controller
115 V, 400 Hz, 30	Voltage present	Director Controller
50 Vdc		
Drive Status		
Train Hydraulic pressure low	Brakes on	Director Controller
Elevation Hydraulic Pressure low	Brakes on	Director Controller
Train Power Amp on	Power on	Director Controller
Elevation Power Amp on	Power on	Director Controller
Train Tripped	(Train Drive Motor Overload. Train Loss of Supply Voltage to Hydraulic Motor. Train Overspeed.)	Director Controller
Elevation Tripped	(Elevation Drive Motor Overload. Elevation Loss of Supply Voltage to Hydraulic Motor. Elevation Overspeed.)	Director Controller
Alarm		
Director Air Supply Off	Blower Off	Director Controller
Electronic Temperature above Normal	Temperature exceeded	Director Controller
Electronic Temperature over Temperature	Temperature exceeded	Director Controller
Interlocks		
Director Start Interlock Open	Interlock open	Director Controller
Director Status		
Director Start	Start Relay picked up	Director Controller
Servo Bias Supplies	Bias Supplies on	Director Controller
Director Start up Complete	Summary-Time delay complete	Director Controller
Director Running	Summary	Director Controller
Director On Line	Summary-Time delay complete mode present	Director Controller

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3.5 Power requirements. The equipment shall operate from sources or primary power that meet the requirements shown below:

a. Alternating current power shall be in accordance with MIL-STD-1399-300 as follows:

115 V	60 Hz	Single Phase	Type I	10 A
115 V	60 Hz	Single Phase	Type I	5 A
440 V	60 Hz	Single Phase	Type I	10 A
440 V	60 Hz	3-Phase	Type I	65 A/phase
115 V	400 Hz	3-Phase	Type III	5 A/phase
440 V	400 Hz	3-Phase	Type I	3 A/phase

b. Direct current power shall have characteristics as follows:

Steady State Voltage:	50 V
Steady State Tolerance:	±2 V
Regulation:	±5 percent
Ripple Amplitude:	2 V root-mean-square (RMS) max

3.6 Electromagnetic compatibility (EMC). Electromagnetic interference (EMI) characteristics of the director (emission and susceptibility) shall be in accordance with MIL-STD-461 (Table A-IV; Note 20; MIL-STD-461A).

3.7 Environmental requirements.

3.7.1 Temperature. All equipment located in enclosed areas shall operate normally within the temperature range of Class 4 Equipment. All equipment exposed to the elements shall operate normally within the Class 2 ambient temperature range as defined in MIL-F-18870.

3.7.2 Humidity. The equipment shall be capable of starting and operating in atmospheres having a relative humidity as great as 95 percent in temperatures up to +40 °C and a constant water content at temperatures of 40 °C and above.

3.7.3 Ship's motion.

3.7.3.1 Normal service conditions. The equipment shall be capable of normal operation under any combination of the following sinusoidal motions together with the maximum normal wind speed.

a. Maximum roll amplitude of ±20 degrees with a period of 9 seconds.
b. Maximum pitch amplitude of ±5 degrees with a period of 6 seconds.
c. Maximum yaw amplitude of ±2.5 degrees with a period of 6 seconds.

The system shall be operable with reduced capability and increased reaction time for roll amplitude up to +30 degrees with the same roll period and the same pitch and yaw motions.

3.7.3.2 Extreme service conditions. The equipment shall not be damaged by the following extreme sinusoidal motion.

a. Maximum roll amplitude of ±45 degrees with a period of 9 seconds.
b. Maximum pitch amplitude of ±15 degrees with a period of 6 seconds.
c. Maximum yaw amplitude of ±7.5 degrees with a period of 6 seconds.

The extreme positions, velocities and accelerations shall not be assumed to occur simultaneously about any of the three axes, but the extreme condition about any one axis may occur simultaneously with the maximum normal condition about the other two axes. Recovery to normal operations shall follow upon return to normal service conditions.

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3.7.4 Mechanical noise.

3.7.4.1 Enclosed area airborne noise. Equipment in enclosed areas shall not generate airborne noise in excess of that permitted for Grade C equipment in MIL-STD-740

3.7.4.2 Structure-borne noise. *The equipment shall fall within the requirements of* Type 3 in accordance with MIL-STD-740.

3.7.5 Wind loading. Wind loading requirements shall be in accordance with MIL-F-18870.

3.7.6 Ice loading. Portions of the equipment exposed to the elements shall start and operate normally when subjected to conditions which would produce a maximum ice load of 4.5 pounds per square foot (lb/ft^2). At points *of limited clearance (1/8 inch or less)* this requirement shall apply after the equipment has broken free. Parts normally covered by a tarpaulin during periods of heavy ice accumulation are excluded from this requirement. A means for removing ice may be included to satify this requirement.

3.7.7 Combination loading. Normal operation of equipment *exposed to the elements shall be possible under* maximum wind and ice loading specified herein, except that maximum acceleration specified in 3.2.2 and frequency response tolerance band in 3.2.2.1 does not apply. The director shall meet the servo system duty "torque/time" requirements as specified in 3.2.5.

3.7.8 Nuclear air blast. Equipment exposed to the elements shall be capable of withstanding 10 pounds per square inch (lb/in^2) over-pressure with a positive phase duration of 1.3 seconds without significant impairment of function.

3.7.9 Shock loading. The equipment shall withstand shock loading of Grade A, deck-mounted, Class I equipment in accordance with MIL-S-901. Performance degradation during shock application is permissible provided normal operation may be resumed by the replacement of minor parts.

3.7.10 Vibration. The equipment shall withstand Type I vibration requirements as specified in MIL-STD-167-1 *except that the frequency of vibrations shall be from 5 to 25 Hz.*

3.7.11 Corrosion resistance. Corrosion resistance of ferrous alloys used in construction of the director shall be in accordance with Requirement 15 of MIL-STD-454. For other materials, Requirement 16 of MIL-STD-454 shall apply as for Class 2 and 3 equipments defined therein.

3.8 Reliability. The director shall have a mean-time-between-failure (MTBF) not less than 1800 hours.

3.9 Maintainability. The director shall have a mean-time-to-repair (MTTR) not greater than 2 hours.

3.10 Government loaned equipment. Government loaned equipment shall be as specified in the contract or order (see 6.2).

3.11 Government furnished equipment. Government furnished equipment shall be as specified in the contract or order (see 6.2).

3.12 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* **4.1.1 Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Each director offered for acceptance under contract shall be subjected to the examinations and tests specified in 4.2.1, 4.2.2, and 4.2.4. In addition, samples selected from units having satisfactorily passed these inspections shall be subjected to the tests of 4.2.5 and 4.2.6 at an activity designated by the contracting officer. Sampling for the tests of 4.2.5 and 4.2.6 shall be as determined by the contract or order (see 6.2). Sampling for verification of reliability and maintainability as specified in 4.2.7 and 4.2.8 shall be as determined by the contract or order. Failure to meet any inspection requirement shall be considered cause for rejection. Acceptance of production units will be based on satisfactory performance of the samples to the test as required by the contract or order.

4.2.1 Examinations. The director shall be subjected to examinations to determine conformance with requirements set forth herein including:

a. parts, materials, and processes
b. cabling and wiring
c. accessibility
d. finish and color
e. identification plates
f. test facilities (test points)
g. operating controls and indicators
h. workmanship
i. marking
j. packaging and packing
k. weights
l. dimensions
m. construction

(Examinations are those inspections which can be made by visual, tactile, or mechanical means to verify that requirements have been met.)

4.2.2 Performance tests. The director shall be subjected to tests to determine conformance with requirements set forth herein. These tests shall be arranged to correlate with the specific performance requirements set forth herein, including tests for:

a. modes of operation
b. train and elevation power drive servo requirements
c. overspeed limits
d. position limits
e. servo system duty*
f. data transmission
g. interface

*Vendor's drive motor data from torque/speed/temperature tests may be used.

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When specified in the contract or order (see 6.2), additional performance tests shall be conducted, including tests for:

a. antenna group inertia
b. antenna group center of gravity
c. director moment of inertia
d. frictional loads.

(Tests are those inspections which can be made by applying predetermined stimuli to the item under test and observing, measuring, or comparing evoked responses to ascertain that a desired performance has been obtained.)

4.2.3 Inspection conditions.

4.2.3.1 Standard conditions. Unless otherwise specified, all inspections shall be made at prevailing conditions of temperature, atmosphere's pressure, and humidity.

4.2.3.2 Attitude. Unless otherwise specified, the equipment shall be tested in its normal operating position.

4.2.3.3 Measurements. During all tests, the accuracy of instruments and test equipment used to control or monitor test parameters shall be verified periodically to the satisfaction of the contracting activity. All instruments and test equipment used in conducting tests specified herein shall:

a. Conform to laboratory standards where calibration is traceable to the prime standards at the National Institute for Standards and Technology.
b. Have an accuracy of at least one-fifth the tolerance for the variable to be measured.

4.2.4 Packaging and packing inspection. Inspection shall be conducted in accordance with MIL-E-17555 to assure compliance with packaging requirements specified herein.

4.2.5 Environmental tests.

4.2.5.1 Nonoperating temperature test. The director shall be subjected to the nonoperating conditions temperature test specified in MIL-F-18870.

4.2.5.2 Operating temperature test. The director shall be subjected to the operating temperature tests of MIL-F-18870 for the appropriate class of equipment as defined herein.

4.2.5.3 Humidity test. The director shall be subjected to humidity tests as specified in MIL-F-18870.

4.2.5.4 Inclination test. The director shall be subjected to the inclination test set forth in MIL-F-18870.

4.2.5.5 Vibration test. The director shall be subjected to Type I vibration tests in accordance with MIL-STD-167-1.

4.2.5.6 Shock test. The director shall be subjected to shock tests for Grade A, Class I equipment as set forth in MIL-S-901.

4.2.5.7 Degree of enclosure. An enclosure test shall be conducted in accordance with MIL-STD-108 to determine conformance with enclosure requirements set forth herein.

4.2.5.8 Mechanical noise tests. Mechanical noise tests shall be conducted in accordance with MIL-STD-740 to determine conformance with the requirements of this specification.

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4.2.5.9 Salt spray tests. A salt spray test in accordance with FED-STD-151 test method 811 shall be conducted on all parts with coatings or finishes, and to frames and enclosure structures assembled for final use. The test shall not be conducted on complete sets or major units. Sample corner structures and other critical sections shall be used for the test. Duration of the tests shall be 200 hours for frame and enclosure structures and 48 hours for individual parts.

4.2.5.10 Icing and loading. Icing and loading tests shall be conducted with the radar antenna mounted on the director.

4.2.5.10.1 Icing. The ambient temperature shall be reduced to −10 °C and a fine spray of water shall be so directed that the junctions between all exposed moving and fixed parts are covered with a 1/2 inch layer of ice. The equipment, including any seal heater, shall be energized and left in the standby condition for 15 minutes. The rotatable parts shall then be energized and resultant operation recorded. With the rotatable parts operating normally, fine spray shall be directed at the junctions of fixed and movable parts and resultant operation recorded. The equipment shall operate normally during this test to be considered acceptable except that changes in the RF propagation characteristics of the antenna due to ice shall not be cause for rejection.

4.2.5.10.2 Wind loading. Tests shall be conducted to determine that the parts of the director exposed to the elements meet the wind loading requirements specified herein.

4.2.5.10.3 Combination loading. Tests shall be conducted to determine that the parts of the director exposed to the elements meet the combined loading requirements specified herein.

4.2.5.10.4 Nuclear air blast. The director shall be tested for satisfactory operation under the power system variations characterizing Type I power specified herein. This test may be conducted during the performance of any or all of the tests of this specification at the discretion of the testing activity or as specified by the contract or order (see 6.2).

4.2.6 EMC tests. The director shall be tested to verify compliance with MIL-STD-461.

4.2.7 Reliability. When specified in the contract or order (see 6.2), a realiability development/growth test shall be conducted in accordance with MIL-STD-781 except that vibration shall not be performed.

4.2.7.1 Definition of failure. A failure is defined as the inability of the director to follow order signals defined herein.

4.2.7.2 Sources of failure data. MTBF predictions shall be based on failure rate data from MIL-HDBK-217. For parts not covered in MIL-HDBK-217 existing data shall be used. The source of the existing data shall be specified and substantiated.

* **4.2.7.3 Burn-in tests.** When specified in the contract or purchase order (see 6.2), a 24-hour burn-in test shall be performed.

4.2.8 Maintainability. When specified in the contract or order (see 6.2), maintainability verification shall be in accordance with MIL-STD-471.

4.2.8.1 Sources of maintainability data. Maintainability predictions shall be based on data given in MIL-HDBK-472.

5. PACKAGING

5.1 Preservation and packaging. The director shall be preserved and packaged in accordance with Level B of MIL-STD-794.

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5.1.1 Unit packaging. Unit packaging of the director set shall be in accordance with Method IIa of MIL-P-116 for equipment installed below decks. For above deck equipment Method III of MIL-P-116 shall be used. Each unit package shall contain one section of the director individually packaged. Gasket material shall be in accordance with MIL-G-12803, Type I, cemented in place with adhesive conforming to MMM-A-1617, Type III. The humidity indicator shall be in accordance with MIL-I-26860, Type II. The rough handling tests of MIL-P-116 shall not apply.

5.1.2 Cushioning. The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices. At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.2 Packing. The director shall be packed in accordance with Level C of MIL-STD-2073-1. The director shall be packed or skidded to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use, and to permit ready handling by overhead lifting devices at that activity. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking. Shipping containers shall be marked in accordance with MIL-STD-129, including serial numbers. Pertinent precautionary markings necessary for full protection of the item packed shall be prominently located as specified in MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The equipment covered by this specification, when installed aboard ship and integrated with other equipments, is intended for use in the fire control of TARTAR guided missiles and, if required, gun control.

* **6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, identification number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
c. Detailed interface and interconnection summary for Fire Control System Mk 74 (see 3.2.7).
d. Government loaned equipment (see 3.11).
e. Government furnished equipment (see 3.12).
f. Equipment to be subjected to the inspections of 4.2.6, 4.2.7, and 4.2.8 (see 4.2).
g. Exceptions to contractor required inspection (see 4.1).
h. Additional performance tests to be conducted (see 4.2.2).
i. Sampling for tests of 4.2.5, 4.2.6, 4.2.7, and 4.2.8 (see 4.2).
j. If nuclear air blast test is to be conducted during other specified tests (see 4.2.5.10.4).
k. Verification of reliability requirements (see 4.2.7).
l. When 24-hour burn-in test shall be performed (see 4.2.7.3).
m. Verification of maintainability requirements (see 4.2.8).

* **6.3 Subject term (key word) listing.**

Fire control
Radar set AN/SPG-51D
Search pattern

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* **6.4 Changes from previous issue.** The margins of this specification are marked with asterisks to indicate where changes have been made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-D-82586B(OS)	2. DOCUMENT DATE (YYMMDD) 31 May 1990

3. DOCUMENT TITLE
Director, Gun and Guided Missile, Mk 73 Mod 2 and Associated Equipment

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commanding Officer Naval Ordnance Station	b. TELEPHONE (*Include Area Code*) (1) Commercial (301)743-4510	(2) AUTOVON 364-4510
ADDRESS (*Include Zip Code*) Code 3730 Indian Head, MD 20640-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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